

# DBDOME — Server Requirements

Requirements to install and run the **DBDOME** database-monitoring platform on a single Windows server. Derived from the installer ( `setup.py` ), the service definitions, and the runtime configuration.

Legend: **Required** = must be present/true to run. **Recommended** = sizing guidance not enforced by the software (tune to your monitored-database count).

## 1. Overview

DBDOME installs to `C:\ProgramData\DBDOME` and runs three logical components as Windows services on one host:

| Component                     | Process / service                                                                                        | Port        |
|-------------------------------|----------------------------------------------------------------------------------------------------------|-------------|
| Web / API (FastAPI + Uvicorn) | <code>dbdome_service.exe</code> → <code>DBDOME_web</code> (or combined <code>DBDOME_dbanalytics</code> ) | <b>8080</b> |
| Scheduler (metric collection) | <code>dbdome_service.exe</code> → <code>DBDOME_scheduler</code>                                          | —           |
| Grafana dashboards            | <code>dbdomeg-server.exe</code> → <code>DBDOME_Grafana</code> (NSSM-wrapped)                             | <b>3000</b> |
| PostgreSQL (analytics store)  | <code>postgresql-x64-18</code>                                                                           | <b>5432</b> |

It connects **out** to the customer's database servers (SQL Server, Oracle, etc.) to collect monitoring metrics.

## 2. Hardware

| Resource | Requirement    | Notes                                                                                                                                                     |
|----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPU      | <b>4 cores</b> | Scheduler runs many concurrent metric queries.                                                                                                            |
| RAM      | <b>16 GB</b>   | PostgreSQL <code>shared_buffers=512MB</code> + up to <code>max_connections=1000</code> ; the embedding model (all-MiniLM-L6-v2) and Grafana add overhead. |
| Disk     | <b>250 GB</b>  | Install ~ a few GB; PostgreSQL data + logs grow with history/retention; allow room for backups. SSD/NVMe recommended for PostgreSQL write throughput.     |

Scale RAM/CPU with the number of monitored databases and metric frequency (M1/M5/M15 collection intervals) if you exceed a typical deployment.

### 3. Operating system

- **Required:** Windows Server (x64). Validated on **Windows Server 2022 / 2025**.
- **Required:** Installer and updater must run **as Administrator** ( `setup.py` aborts otherwise — it installs services, writes firewall rules, configures PostgreSQL).
- **Recommended:** Latest Windows updates; stable hostname; static IP / reserved DHCP.

### 4. Software prerequisites

#### Bundled by the installer (no manual install)

These ship in the install package and are installed/configured by `setup.py` :

- **PostgreSQL 18.3** ( `postgresql-18.3-3-windows-x64.exe` ) → `C:\Program Files\PostgreSQL\18`
- **Microsoft ODBC Driver 17/18 for SQL Server** ( `msodbcsql.msi` ) — for SQL Server monitoring
- **NSSM** ( `nssm.exe` ) — wraps `dbdomeg-server.exe` (Grafana) as a Windows service
- **DBDOME binaries** — `dbdome_service.exe` , `dbdome_dbanalytics.exe` , `dbdomeg-server.exe` , `grafana.exe` (PyInstaller-frozen; Python runtime is embedded — **no separate Python install required**)
- **Embedding model** — `all-MiniLM-L6-v2` (local; referenced by `.env` `LLM=` )

#### Must exist / be reachable

- **Oracle Instant Client** — only if monitoring Oracle. Path set via `.env` `ORACLE_CLIENT_LIB_DIR` (e.g. `C:\oracle\instantclient_23_0` ).
- **DVD / mounted media or local source** containing the install bundle ( `bin/` , `postgres/install/` , `data/` ).

### 5. Network & ports

#### Inbound (must be open to dashboard/API users)

| Port | Protocol | Service        | Purpose                                                |
|------|----------|----------------|--------------------------------------------------------|
| 8080 | TCP      | DBDOME Web/API | Web UI & REST endpoints                                |
| 3000 | TCP      | Grafana        | Dashboards                                             |
| 5432 | TCP      | PostgreSQL     | Analytics DB (open only if remote DB access is needed) |

The installer creates these firewall rules (profiles: **domain, private**): `DBDOME HTTP Server` (8080), `DBDOME Grafana` (3000), `DBDOME PostgreSQL` (5432).

#### Outbound (to monitored databases)

The server must reach each monitored DB on its listener port, e.g.: - **SQL Server** — TCP **1433** (or named-instance port) - **Oracle** — TCP **1521** - **PostgreSQL/MySQL/etc.** — their respective ports

## IP / addressing

- The server's **local LAN IPv4** is written to `.env` `ORG_IP` and used to build dashboard/redirect URLs. Clients must be able to reach the server at that address.
- A static/reserved IP is strongly recommended (URL links break if it changes).

## 6. Database (PostgreSQL) configuration

Applied by the installer:

- **Instance:** PostgreSQL 18 on port **5432**, service `postgresql-x64-18`.
- **Database:** `dbanalytics`.
- **Roles:** `postgres`, `dbdome_mon_usr`, `dbdome_adm`.
- `pg_hba.conf`: `trust` for `127.0.0.1/32` and `:::1/128` (local-only).
- `postgresql.conf` **tuning:**
  - `max_connections` = 1000
  - `shared_buffers` = 512MB
  - `logging_collector` = off
- **Restore:** baseline `dbanalytics_install.backup` restored on first install.

`max_connections` = 1000 is high — ensure RAM is sized accordingly, or front it with a pooler for very large deployments.

## 7. Disk layout

| Path                                         | Contents                                                                                                           |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>C:\ProgramData\DBDOME\</code>          | Install root ( <code>.env</code> , <code>install.log</code> , <code>update.log</code> )                            |
| <code>C:\ProgramData\DBDOME\bin\</code>      | Service exes, <code>nssm.exe</code> , Grafana, <code>icons\</code> , <code>.env</code> (the one the services read) |
| <code>C:\ProgramData\DBDOME\conf\</code>     | Grafana provisioning / <code>custom.ini</code>                                                                     |
| <code>C:\ProgramData\DBDOME\data\</code>     | <code>grafana.db</code> (~20 MB)                                                                                   |
| <code>C:\Program Files\PostgreSQL\18\</code> | PostgreSQL binaries + <code>data\</code>                                                                           |

Public desktop shortcut `DBDOME.lnk` → `http://localhost:8080` (icon `C:\ProgramData\DBDOME\bin\icons\dbdome.ico`).

## 8. Windows services

| Service                                                         | Startup   | Recovery                                |
|-----------------------------------------------------------------|-----------|-----------------------------------------|
| DBDOME_web / DBDOME_scheduler (or combined DBDOME_dbanalytics ) | Automatic | Auto-restart on failure (3×, 5 s apart) |
| DBDOME_Grafana (NSSM)                                           | Automatic | Auto-restart on exit                    |
| postgresql-x64-18                                               | Automatic | Windows default                         |

All services run with privileges sufficient to read `C:\ProgramData\DBDOME` and bind their ports.

## 9. Security considerations

- Restrict inbound 8080/3000/5432 to trusted networks (firewall profiles are domain/private by default — review for your environment).
- PostgreSQL uses local `trust` auth; do **not** expose 5432 publicly without changing auth.
- Monitored-database credentials are stored in the analytics DB — protect the host and DB access accordingly.
- Change default service/DB passwords before production use.

## 10. Pre-installation checklist

- ☐ Windows Server x64, fully updated
- ☐ Administrator account to run the installer
- ☐ Static/reserved IPv4; firewall manageable
- ☐ Hardware per §2 (CPU/RAM/SSD)
- ☐ Install media/bundle available (DVD `DBDOME_DVD_ROM` or local path)
- ☐ Outbound network access to every monitored DB (host:port)
- ☐ Oracle Instant Client present (only if monitoring Oracle)
- ☐ Ports 8080 / 3000 / 5432 free (not used by other software)

Source of truth: `C:\dev\dbdome_setup\setup.py` (installer) and the DBDOME service/runtime configuration.